

Algorithmic Non-Determinism

Next Token: Cat (50%), Dog (30%), Bird (20%)

- Softmax
- Temperature
 - Top-k
 - Top-p
- Greedy Decoding ($T=0$)

Hardware Non-Determinism

Meaning: Even without random sampling, hardware and numerical differences can cause output variation

Downstream Mitigation

Meaning: We design applications to handle potentially inconsistent or invalid LLM outputs

- JSON Schema
- Error Handling
- Validation